Networks 2 (TEL 418) Project Report: WiFi Doctor

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1 Introduction

Ensuring optimal Wi-Fi performance remains a challenge due to numerous factors, including network congestion, interference, and device configurations. To address these challenges, we made the "Wi-Fi Doctor", as a project for class Networks 2 (TEL 418) at TUC, a system designed to monitor and analyze Wi-Fi network performance at the device side, identifying potential bottlenecks.

The Wi-Fi Doctor focuses exclusively on the diagnostic aspects of Wi-Fi performance, utilizing packet capture (pcap) files to extract relevant metrics and evaluate network conditions. The system consists of several key components: a Wi-Fi sniffer that collects network traffic data, a pcap parser that extracts performance-related parameters, a performance monitor that evaluates network conditions, and a performance analyzer that determines the underlying causes of network issues. The system also includes a visualizer to present the collected data in an intuitive format.

This project is divided into two main parts. The first part involves Wi-Fi network density by analyzing beacon frames across different network environments and frequency channels. The second part focuses on estimating theoretical downlink throughput and identifying factors that influence performance degradation. By leveraging key metrics such as signal strength, data rates, and interference levels, the Wi-Fi Doctor provides valuable insights into network behavior.

The results will highlight key network density metrics, as well as the primary factors affecting throughput.

2 Design

An overview of Wi-Fi Doctor system is shown in the figures 1 and 2 below. The program starts by giving the user the choice to perform density analysis or performance analysis and then proceeds by parsing a pcap file, saved from wireshark, then extracting the necessary fields from each packet and perform data analytics regarding wifi density metrics or performance indicators. Finally the visualizer displays the results in plots, which helps us understand the behavior of the network.

Figure 1: Block diagram of density analysis

- **Parser:** The Parser takes a .pcap (.pcapng) file as an input and extracts the following fields using the library *pyshark*, **{BSSID,Transmitter MAC, Receiver MAC address, Type/subtype, PHY Type, MCS Index, Bandwidth, Spatial Streams, Short GI, Data rate, Channel, Frequency, Signal strength, Signal/noise ratio, TSF Timestamp}**. If the selected option was *density analysis* the parser applies a filter to keep only the beacon frames.

Figure 2: Block diagram of performance analysis

- **Monitor:** The Monitor takes a python dictionary as input and transforms it to a dataframe using the library *pandas*. The pandas library provides functions that are able to perform wide and narrow operations. The dataframe contains certain fields at this part and calculates: {*The number of unique transmitter MACs per BSSID, The total packets transmitted per BSSID, The total packets transmitted per PHY type, The total packets transmitted per channel, The total packets transmitted per frequency band, The RSSID for the entire dataset, The total number of SSIDs, The mean RSSI per BSSID*}. Also, when it comes to the density analysis, the monitor also normalizes the metrics across the time axis, i.e. dividing the metrics by the duration of the capture. That way, the analysis is independent of the capture duration.

  For the performance analysis, the monitor also calculates the throughput (eq.2) as described in project description [6].

- **Analyzer:** The Analyzer calculates the loss rate, the mean throughput and adds additional fields to the original Dataframe: **Expected Data Rate** value for each record containing {*mcs index, bandwidth, phy type, spatial streams, short gi*} from the table [3].

$$\textbf{Rate Gap} := \text{Expected Data Rate} - \text{Data Rate} \tag{1}$$

$$\textbf{Throughput} := \text{Data Rate} \cdot (1 - \text{Loss Rate}) \tag{2}$$

- **Visualizer:** Visualizer creates and displays all necessary figures including plots and histograms for every density metric and performance indicator. Also, it calculates the rolling average for a window (e.g. 300 values) for better and smoother plots.

3 Evaluation

In this section we analyze the results produced by the WiFi Doctor. First, we explain the density metrics. Then we seek to identify the performance bottlenecks of the network and explain the **why** these factors impact the performance.

3.1 WiFi Network Density

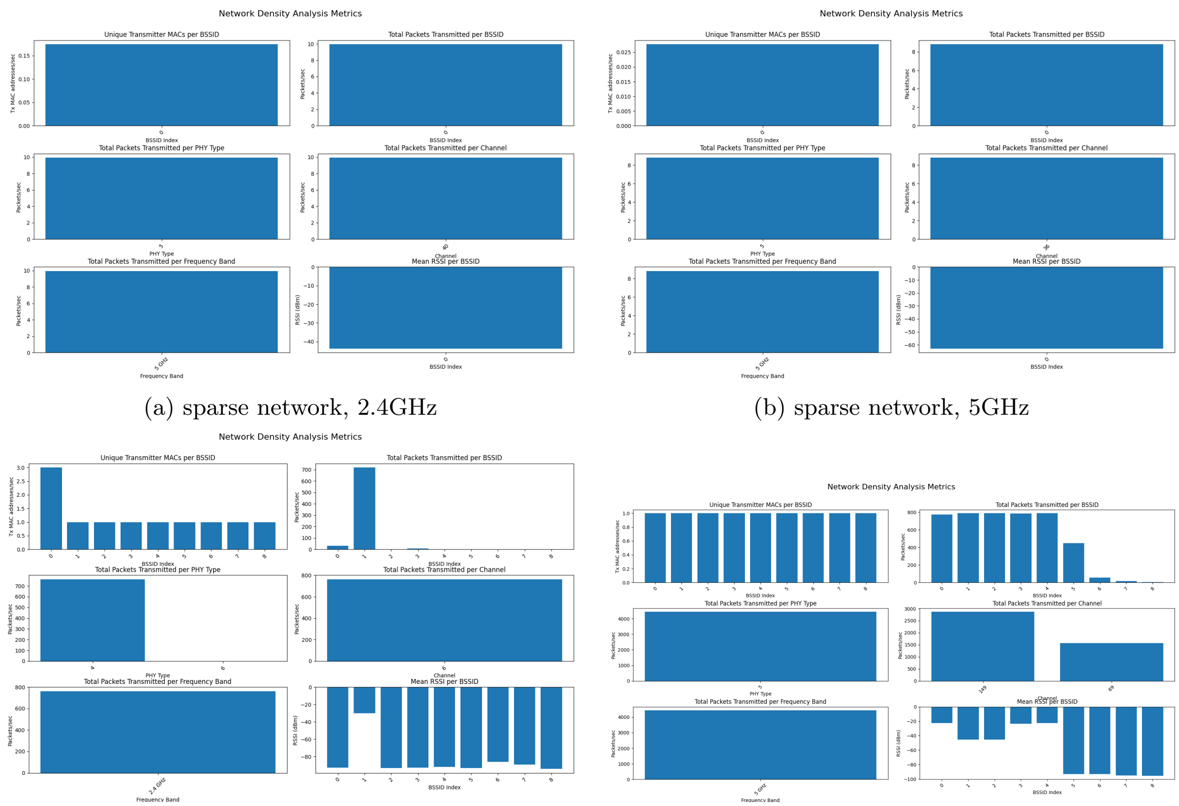
In order to identify the network activity and address the it's density we used the following metrics,

- **Packets/sec per BSSID**: is the packet rate for each unique BSSID (Basic Service Set ID). Shows the traffic distribution across different APs and identifies overloaded or underutilized APs.

- **Packets/sec per PHY Type**: is the packet rate by physical layer technology (802.11a/b/g/n/ac/ax). Shows THE technology mixture in the environment. It's important because the newer legacy devices handle better the dense environments.

- **Packets/sec per channel**: shows the traffic distribution across different channels. Shows channel utilization patterns and identifies congested channels. Reveals interference patterns, because overlapping channels may cause interference.

- **Packets/sec per Frequency Band**: the traffic split between 2.4GHz and 5GHz (and 6GHz if available). 2.4GHz is typically more crowded but better range.



(a) sparse network, 2.4GHz

(b) sparse network, 5GHz

(c) Dense network (pcap file from lab 1), 2.4GHz

(d) Dense network (pcap file from [4]), 5GHz

Figure 3: Density metrics

- **Mean RSSI per BSSID**: is the average Received Signal Strength Indicator for each AP. This metric shows coverage patterns.
- **Tx MACs/sec**: This metric is more useful when working not only with beacon frames. In beacon frames (usally), Tx MAC = BSSID.

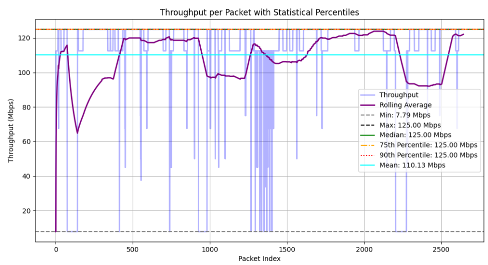
For the sparse networks we took traces from our home network where we had a single access point thus a single BSSID. For dense networks for channel 2.4 GHZ we used the given file from lab 1 **trace 801_11.pcapng** and for channel 5 GHZ we downloaded the trace **ATM_U7Pro_iPhone16_MLO_3.pcapng** from [4].
It is observed that in the dense networks there are more bssids (9 bssids) that contain traffic from different mac addresses. Another indicator of a dense network is the number of *Packets/sec per BSSID*. In the dense networks the packets are distributed more across the available bssids, especially in the 5 GHZ network the packets are distributed evenly between 6 bssids.
Additionally the *Mean RSSI per BSSID* shows the correlation between the signal strength and the BSSIDs. The BSSIDs that have the majority of total packets have a higher RSSI value than the rest. In fig 3c the BSSID with the higher RSSI has the most packets, which means it was probably the closest access point to the device that captured the trace that's why it detected most of the packets. In fig 3d there are 5 BSSIDs with high RSSI (-20 and -40) that have the most packets transmitted through them. There are also 4 BSSIDs with bad RSSI, so we can assume they were far away from the "sniffer" device that's why the traffic is so low and the packets limited.

3.2 WiFI Network Performance

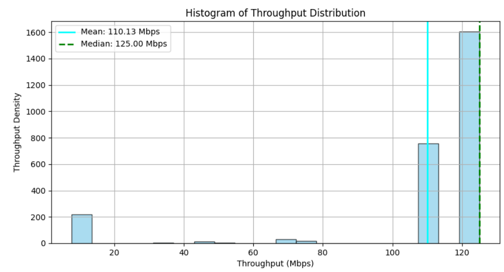
For this part the Parser filtered only the 802.11 data frames of the given file *HowIWiFi_PCAP.pcap*, while the performance monitor analyzed the traffic from the **AP (2C:F8:9B:DD:06:A0)** to the **device (00:20:A6:FC:B0:36)**. In order to check the quality of the network, we calculated the theoretical throughput and compared it to other metrics **{PHY Type, Bandwidth, Short GI, Data**

Rate, MCS Index, Signal strength, Rate Gap} to identify the performance bottlenecks.

The formula for the theoretical throughput is given by eq.2. The loss rate was configured by calculating the retry flag [2]. The Throughput is represented with in 5a. The time-series and the distribution analysis used the data rate values and the mean value to produce the plots. In fig 5a The rolling average of the throughput is proportional with the rolling average of the data rate and inversely proportional to the rolling average of the loss rate, thus the eq 2 is confirmed.

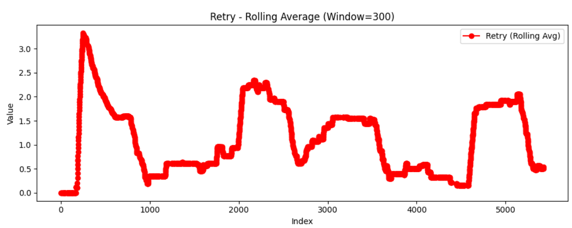


(a)

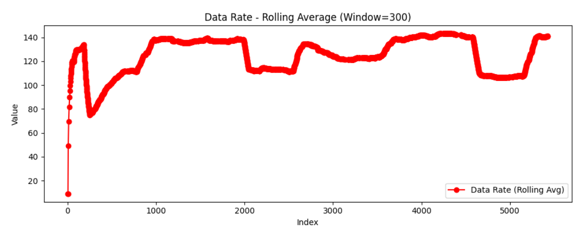


(b)

Figure 4: a)Time-Series: Throughput value, Rolling Average, Min value, Max value, Median, 75th percentile, 90th percentile, Mean value. b)Distribution: Total packets per throughput value, mean and median



(a) Rolling average losses (raw data takes boolean values but this is an average of 300 entries)



(b) Rolling average data rate

Figure 5: Performance metrics

The behavior of the Data Rate can be explained by evaluating the following fields to make some assumptions.

- Wi-Fi configuration (PHY Type, Bandwidth, Short GI)
- Wireless channel (MCS Index, Signal Strength)
- Wireless channel / interference

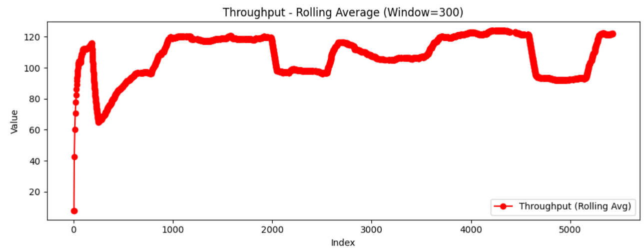
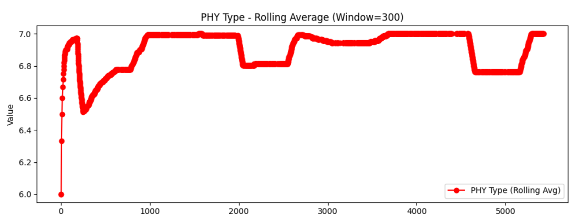


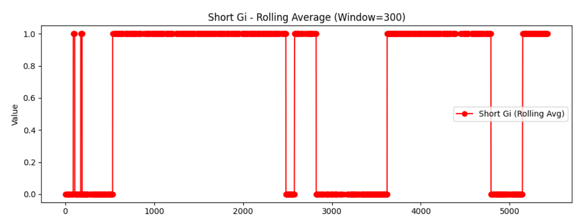
Figure 6: Throughput rolling average (window = 300)

3.2.1 Wi-Fi configuration

The bandwidth values are equal to 20 Mhz throughout the capture so it doesn't affect the data rate. The phy type values are either 6 (802.11g) or 7 (802.11n) [7], in the areas close to 7 the throughput is increased because there are more 802.11n packets otherwise there is a majority of 802.11g packets and the throughput is decreased. This happens because the 802.11n is newer (2009) that the 802.11g (2003), it supports maximum link rates from 6.5 to 600 mbps, while 802.11g 6.5 to 54 mbps, as a result it makes it easier to achive better throughput. Moreover according to [5] the shorter guard interval results in a higher packet error rate when the delay, spread of the channel, exceeds the guard interval or if timing synchronization between the transmitter and receiver is not precise. From fig 7b it is observed that when the short gi is equal to 0 the throughput values are in a local minimum. Taking both figures 7a 7b into account we see that the rolling average of the phy type is almost identical to the throughput's and they are correlated more.



(a) Rolling average PHY Type

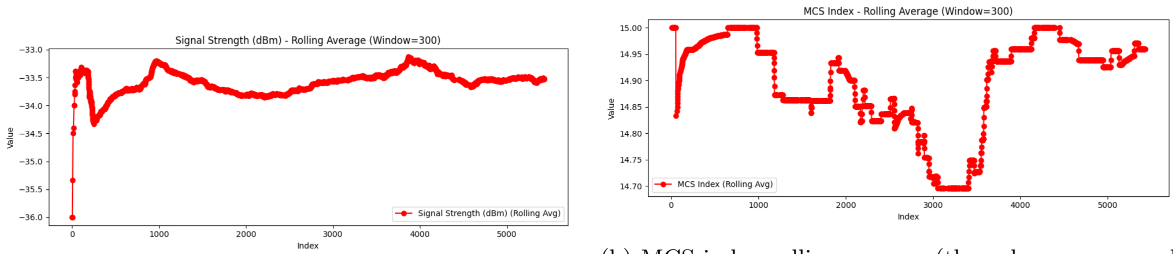


(b) Rolling average and approximation for short gi

Figure 7: Wi-Fi configuration metrics

3.2.2 Wireless channel

Good signal strength (and signal quality - SNR) allows the system to chose higher MCS index which corresponds to better throughput. This happens because, higher MCS index allows the use of more complex modulations and better coding rates. Small differences between fig. 8a and 8b could occur due to the fact that the system is selecting MCS based on SNR (not RSSI), RSSI is an approximation of SNR without considering the noise factor.



(a) Signal strength rolling average

(b) MCS index rolling average (the values correspond to the average of discrete values)

Figure 8

3.2.3 Interference

The packets provided by the pcap don't contain any noise/interference information, so we are using table 1 from [1] to identify potential interference which cause poor throughput. From equation 1 and table 1 we see that when *Rate Gap* is negative this means that the difference in the expected rate gap and the actual rate gap could be caused by the interference. In figure 9 is the timeseries of the rate gap. We can't see any direct correlation with throughput of fig. 5a but the interference affects the overall quality of the network and the throughput.

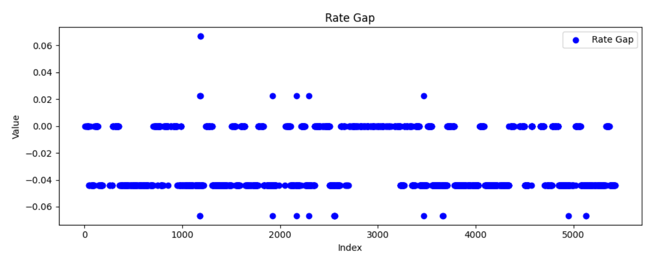


Figure 9: Rate Gap per packet index

PHY Rate/ Exp. PHY Rate (R_E)	HIGH	LOW
HIGH	good performance	poor performance (RA dynamics)
LOW	poor performance (interference/RA dynamics)	poor performance (poor coverage)

Table 1: Table from [1] for the interpretation of rate gap.

4 Related Work

A lot of the metrics used in this work have extensively studied in I.Pefkianakis et al. [1]. They characterized home networks using data gathered from getaways. They introduced the expected gap and rate gap metrics in order to estimate the interference in the network.

References

[1] Ioannis Pefkianakis et al. "Characterizing home wireless performance: The gateway view". In: *2015 IEEE Conference on Computer Communications (INFOCOM)*. 2015, pp. 2713–2731. DOI: 10.1109/INFOCOM.2015.7218663.

[2] *Loss Rate Using Wireshark*. Accessed: March 2025. URL: https://semfionetworks.com/blog/calculate-802-11-retry-rate-with-wireshark/.

[3] *MCS index table*. Accessed: March 2025. URL: https://www.watchguard.com/help/docs/help-center/en-US/Content/en-US/WG-Cloud/Devices/access_point/deployment_guide/Checklist_Charts/mcs_80211n_ac.pdf.

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[6] *Wi-Fi Doctor Project Description*. TUC-Eclass.

[7] *Wireshark source code*. URL: https://gitlab.com/wireshark/wireshark/-/blob/master/epan/dissectors/packet-ieee80211-radio.c.